Today

- Position and momentum
- Expectation value
- The uncertainty principle

1.11 Position and momentum

Let us focus on a particle in one-dimensional space with coordinate x. Now $|\psi(x,t)|^2$ represents the probability density for finding the particle at point x at time t, and $|\psi(x,t)|^2 dx$ is the probability for finding the particle between x and $x+dx$. Observe that $|\psi(x,t)|^2$ is a real number with unit of $m^{-1/2}$ that can be plotted as a function of x at a given instant of time. Suppose we have a profile of $|\psi(x,t_1)|^2$ shown in the figure below at some time t_1 for a system containing a single particle.

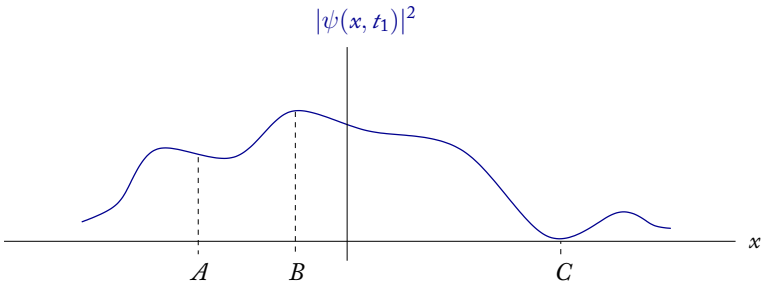


Figure 1.11.1: Probability density at instant of time t_1.

Of all the points on x, the particle is most likely to be found at B, where the probability density is largest, and least likely to be found at C, where the probability density vanishes. The probability for finding the particle anywhere between points A and B at time t_1 is

$$P_{AB}(t_1) = \int_A^B |\psi(x,t_1)|^2 dx, \tag{1.11.1}$$

which represents the area under the curve between the two points. The total probability for finding the particle at any point is $\int_{-\infty}^{\infty} |\psi(x,t_1)|^2 dx = 1$, which is just the total area under the curve.

Now we seek to find the average value of position. We can think of two scenarios. One is preparing a large number of identical systems and measuring the position of the particle for

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each. The other is repreparing the same system again and again and repeating the experiment of measuring the position of the particle. [Note that this is not the same as measuring the location of the same particle again and again without repreparing the system.] In either case, the average value (also called the expectation value) of the position of the particle is

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} x \mathcal{P} dx = \int_{-\infty}^{\infty} x |\psi(x,t)|^2 dx \tag{1.11.2}$$

Question 1.11.1: Which of the following are true about expectation (or average) value? (A). It is the most likely value of a measurement, (B). It is the average value of repeated measurements on identical systems, (C). It may not be an outcome of any single measurement, (D). It is the expected value of a measurement.

Answer: (B) and (C) are the answers. Consider a system with two energy levels of 0 and E_1. If both states are equally likely, the average energy would be $\frac{1}{2}E_1$. However, no single measurement would actually produce $\frac{1}{2}E_1$. The phrase "expectation value" is interchangeable with average value, it is does not mean what is expected to be obtained in a measurement.

Next we want to write an expectation value, $\langle p \rangle$, for the momentum of the particle. Consider

$$\begin{aligned}
\frac{d}{dt}\langle \hat{x} \rangle &= \frac{d}{dt}\int_{-\infty}^{\infty} x|\psi(x,t)|^2 dx = \int_{-\infty}^{\infty} x\frac{\partial \mathcal{P}}{\partial t} dx = -\int_{-\infty}^{\infty} x\frac{\partial \mathcal{J}}{\partial x} dx \\
&= -\int_{-\infty}^{\infty}\left[\frac{\partial}{\partial x}(xJ) - J\right] dx = -(xJ)|_{x=-\infty}^{\infty} + \int_{-\infty}^{\infty} J dx = \int_{-\infty}^{\infty} J dx \\
&= -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\left(\psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x} - \psi(x,t)\frac{\partial \psi(x,t)^*}{\partial x}\right) dx \\
&= -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\left(\psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x} - \frac{\partial}{\partial x}(\psi(x,t)^*\psi(x,t)) + \psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x}\right) dx \\
&= -\frac{i\hbar}{m}\int_{-\infty}^{\infty}\psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x} dx = \frac{1}{m}\int_{-\infty}^{\infty}\psi(x,t)^*\frac{-i\hbar\partial\psi(x,t)}{\partial x} dx.
\end{aligned} \tag{1.11.3}$$

The expectation value of momentum is asserted to be that of the mass of the particle times the rate of change of the expectation value of the position,

$$\langle \hat{p} \rangle \equiv m\frac{d}{dt}\langle \hat{x} \rangle = \int_{-\infty}^{\infty}\psi(x,t)^*\frac{-i\hbar\partial\psi(x,t)}{\partial x} dx = \int_{-\infty}^{\infty}\psi(x,t)^*\hat{p}\psi(x,t) dx, \tag{1.11.4}$$

where, again, $\hat{p} = -i\hbar\frac{\partial}{\partial x}$ is the momentum operator.

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Similarly, we can rewrite the expectation value of position as

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi(x,t)^* \hat{x} \psi(x,t) dx. \tag{1.11.5}$$

Denoting either one of the position or the momentum operator or any other function of position and momentum, and generalizing to three dimensions, by $\hat{O}(\hat{\vec{r}}, \hat{p}) = \hat{O}(\vec{r}, -i\hbar\vec{\nabla})$, the expectation value of $\hat{O}$ is given by

$$\langle \hat{O} \rangle = \int \psi(\vec{r},t)^* \hat{O}(\vec{r}, -i\hbar\vec{\nabla}) \psi(\vec{r},t) d^3\vec{r}, \tag{1.11.6}$$

where the operator is inserted in-between $\psi(\vec{r},t)^*$ and $\psi(\vec{r},t)$

1.12 The uncertainty principle

Let us recall our discussion on double-slit electron diffraction before and after measurement.

Question 1.12.1: The figure below shows $|\psi|^2$ before and after measurement.

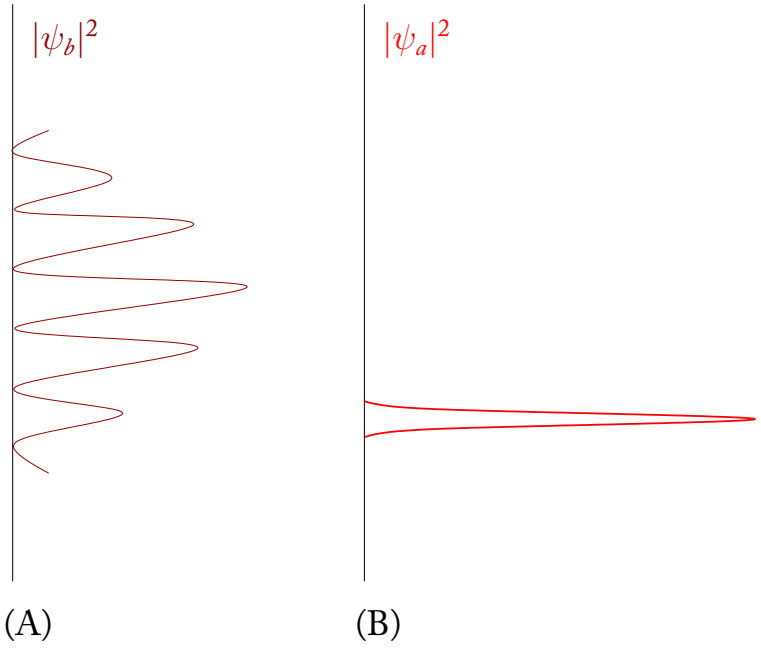


Figure 1.12.1: Probability density before (A) and after (B) measurement.

In which case is the particle's location better determined? What about the momentum?

Answer: In (B), the particle can be located within a narrow width on the screen

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in comparison to (A). On the other hand, the wavelength is better quantified in (A) compared to (B). Because the wavelength is related to linear momentum, the momentum of the particle is better quantified in (A) than in (B).

It turns out, generally, that when the position of the particle is better determined, is momentum is not and vice versa. In order to quantify this uncertainty in measurements of position and momentum, we use the standard deviation for each, Δx and Δp, which is defined as

$$\Delta x = \sqrt{\langle(\hat{x} - \langle\hat{x}\rangle)^2\rangle}, \qquad \Delta p = \sqrt{\langle(\hat{p} - \langle\hat{p}\rangle)^2\rangle} \tag{1.12.1}$$

Using $\hat{O}$ for either one of $\hat{x}$ or $\hat{p}$, we note that

$$(\Delta O)^2 = \langle(\hat{O} - \langle\hat{O}\rangle)^2\rangle = \langle\hat{O}^2 + \langle\hat{O}\rangle^2 - 2\hat{O}\langle\hat{O}\rangle\rangle = \langle\hat{O}^2\rangle - \langle\hat{O}\rangle^2. \tag{1.12.2}$$

The uncertainty principle for position and momentum, which we will investigate further later as the course progresses, states that

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}, \tag{1.12.3}$$

and it quantifies our observation of the uncertainty in position becoming larger when the uncertainty in moments is smaller and vice versa.

To get some physical intuition, let us consider an electron. In order to find its location we have to see it by shining light. The scattering of light from a target has the same pattern as the diffraction from a hole of the same size as the target except in the forward direction. [This is called Babinet's principle in optics.] Therefore, considering a slit of the size of the electron Δx, we have the diffraction pattern shown in the figure below.

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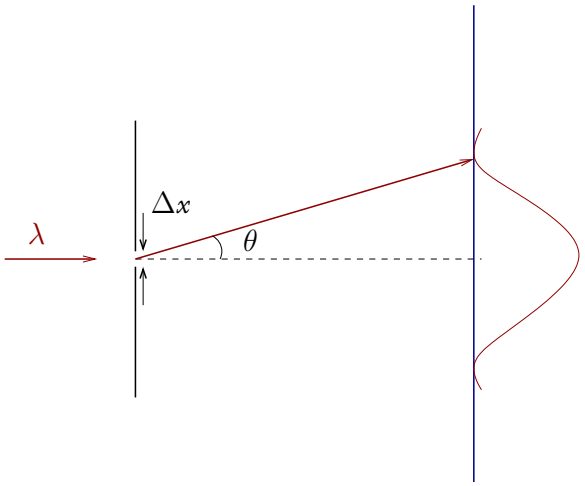


Figure 1.12.2: Scattering of a photon by an electron of size Δx can be viewed as diffraction of a photon by a slit of width Δx.

Because $\sin\theta = \frac{\lambda}{\Delta x}$, θ gets larger as Δx gets smaller. But the component of the magnitude of momentum of the photon along x varies from $p_x = 0$ when it goes straight undeflected to $p_x = \frac{h}{\lambda}\sin\theta$ when it emerges at θ. In other words the range of $\Delta p_x = \frac{h}{\lambda}\sin\theta$ (and the uncertainty in Δp_x) increases with increasing θ, which corresponds to decreasing Δx. Because the total momentum of the photon electron system along x must stay zero, the magnitude of the electron's momentum changes by the same amount. Thus, when Δx is better measured and the electron's position is well known, Δp_x is larger the electron's momentum is more uncertain. This is all because of wave-particle duality. In particular, if Planck's constant were to be zero and the wave property of the electron were not there, both its position and momentum could be measured with arbitrary precision. Therefore, $\Delta p_x = \frac{h}{\lambda}\sin\theta = \frac{h}{\lambda}\frac{\lambda}{\Delta x} = \frac{h}{\Delta x}$, or $\Delta x \Delta p_x = h$, consistent with $\Delta x\,\Delta p \geq \frac{\hbar}{2}$ and that we cannot do better than $\Delta x\,\Delta p = \frac{\hbar}{2}$. The uncertainty is intrinsic to quantum mechanics and not of any specific measuring device or scheme.

Question 1.12.2: Based on our investigation so far what is the reason that leads to the uncertainty principle $\Delta x \Delta p \geq \frac{\hbar}{2}$?

Answer: The wave-particle duality. Because a particle is represented by a wave, and a wave is spread out in space, the position has uncertainty. Similarly, because the wave describe a particle that tends to be localized around the particle, the wavelength (and the momentum) has uncertainty. Or, simply because $\hbar \neq 0$, which is the reason why we have wave-particle duality.

Question 1.12.3: Why is is then that we could measure both the position and

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the momentum of a chalk without difficulty?

Answer: Because $\hbar$ is too small for the chalk, the uncertainty principle does not have meaningful effect. For instance, even if we are interested in locating a chalk to within a micrometer precision, $\Delta x \sim 10^{-6}$m, we still have a large room for Δp to satisfy the uncertainty principle. In fact, even for an electron moving in a macroscopic region, such as in a room or in a cathode ray tube, classical physics is enough, since we are not looking at the electron very closely to within atomic distance scales. The uncertainty in position and momentum become relevant when we try to locate the electron to distance scales comparable to its deBroglie wavelength.

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